

SNOWFLAKE TRAINING

ASSIGNMENT

QUESTION:

Retail Marketing Campaign Analytics

A retail company wants to analyze customer buying behavior and marketing campaign performance using Snowflake sample data.

The company has three teams:

1. Marketing Analysts
2. Finance Team
3. ETL Developers

Each team should use a separate warehouse. The company also wants to improve query performance using:

1. Warehouse scaling
2. Query result cache
3. Cluster key on customer sales date

Use Snowflake sample data from:

SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.STORE_SALES

SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.CUSTOMER

SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.DATE_DIM

SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.ITEM

Questions

Part 1: Warehouse Setup

Create three warehouses:

1. MARKETING_WH for marketing analysts
2. FINANCE_WH for finance team
3. ETL_WH for ETL developers

Use different warehouse sizes based on workload.

Part 2: Role Setup

Create three roles:

1. MARKETING_ANALYST
2. FINANCE_USER
3. ETL_DEVELOPER

Grant each role access only to its assigned warehouse.

Part 3: User Setup

Create users:

1. MKT_USER1
2. FIN_USER1
3. ETL_USER1

Set default role and default warehouse for each user.

Part 4: Create Performance Demo Database

Create a database called:

RETAIL_MARKETING_DB

Create a table called:

CAMPAIGN_SALES

using Snowflake sample tables.

Part 5: Grant Permissions

Grant proper database, schema, and table permissions to each role.

Marketing and Finance can only read data.

ETL Developer can create and modify tables.

Part 6: Performance Testing

Run a sales analysis query using XSMALL warehouse.

Then scale up the warehouse to SMALL.

Compare performance.

Then scale it down again.

Part 7: Cache Testing

Run the same query twice.

Observe why the second query is faster.

Part 8: Cluster Key Testing

Run a date-filter query before clustering.

Add cluster key on SALE_DATE.

Run the same query again.

Compare performance.

STEPS OR PROCESS:

1.WAREHOUSE SETUP

----MARKETING WAREHOUSE

create or replace warehouse marketing_wh

WITH WAREHOUSE_SIZE = 'XSMALL'

WAREHOUSE_TYPE = 'STANDARD'

AUTO_SUSPEND = 60

AUTO_RESUME = TRUE

MIN_CLUSTER_COUNT = 1

MAX_CLUSTER_COUNT = 1;

-----FINANACE WAREHOUSE

create or replace warehouse finance_wh

WITH WAREHOUSE_SIZE = 'XSMALL'

WAREHOUSE_TYPE = 'STANDARD'

AUTO_SUSPEND = 60

AUTO_RESUME = TRUE

MIN_CLUSTER_COUNT = 1

MAX_CLUSTER_COUNT = 1;

-----ETL WAREHOUSE

create or replace warehouse elt_wh

WITH WAREHOUSE_SIZE = 'SMALL'

WAREHOUSE_TYPE = 'STANDARD'

AUTO_SUSPEND = 60

AUTO_RESUME = TRUE

MIN_CLUSTER_COUNT = 1

MAX_CLUSTER_COUNT = 1;

NAME	TYPE	SIZE	STATUS	RUNNING	QUEUED	TABLES
MARKETING_WH	Standard	X-Small	Suspended	0	0	—
ELT_WH	Standard	Small	Suspended	0	0	—
FINANCE_WH	Standard	X-Small	Suspended	0	0	—

2.ROLE SETUP

----MARKET ANALYST ROLE

```
CREATE ROLE marketing_analyst;
```

```
GRANT USAGE ON WAREHOUSE marketing_wh TO ROLE marketing_analyst;
```

-----FINANCE USER ROLE

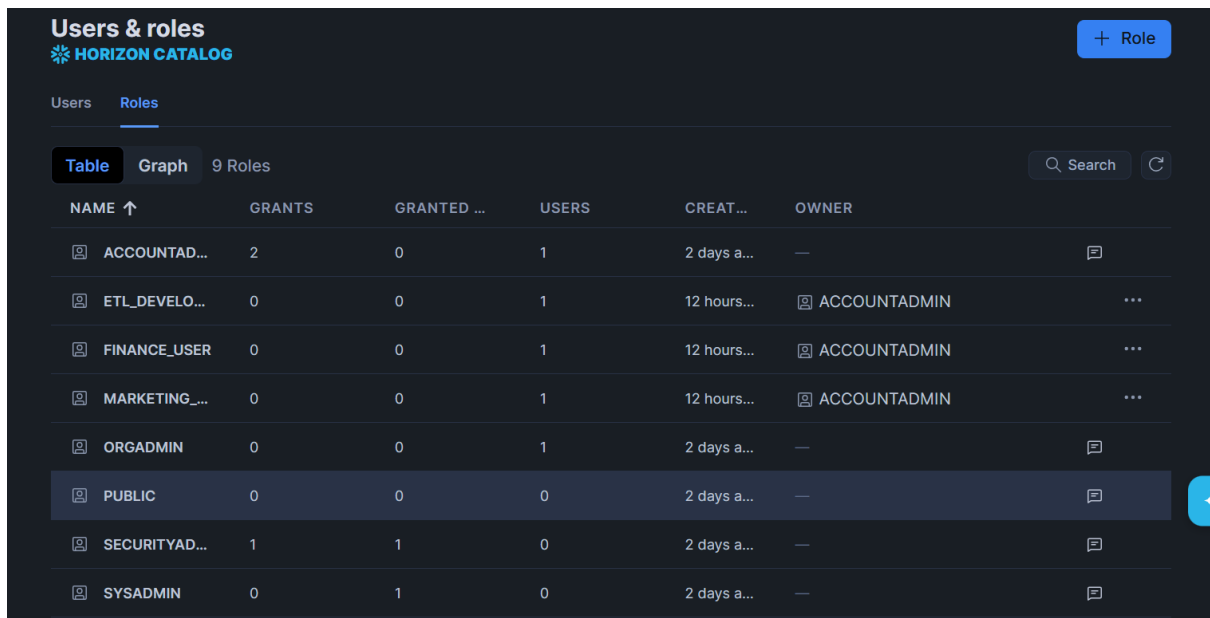
```
CREATE ROLE finance_user;
```

```
GRANT USAGE ON WAREHOUSE finance_wh TO ROLE finance_user;
```

-----ETL DEVELOPER ROLE

```
CREATE ROLE etl_developer;
```

```
GRANT USAGE ON WAREHOUSE elt_wh TO ROLE etl_developer;
```



NAME ↑	GRANTS	GRANTED ...	USERS	CREAT...	OWNER
ACCOUNTADMIN	2	0	1	2 days a...	—
ETL_DEVELOPER	0	0	1	12 hours...	ACCOUNTADMIN
FINANCE_USER	0	0	1	12 hours...	ACCOUNTADMIN
MARKETING_ANALYST	0	0	1	12 hours...	ACCOUNTADMIN
ORGADMIN	0	0	1	2 days a...	—
PUBLIC	0	0	0	2 days a...	—
SECURITYADMIN	1	1	0	2 days a...	—
SYSADMIN	0	1	0	2 days a...	—

3.USER SETUP

----CREATING MKT USER

```
CREATE USER mkt_user1 PASSWORD = 'mkt' LOGIN_NAME = 'mkt_user'  
DEFAULT_ROLE='marketing_analyst' DEFAULT_WAREHOUSE = 'marketing_wh'  
MUST_CHANGE_PASSWORD = FALSE;
```

---CREATING FIN USER

```
CREATE USER fin_user1 PASSWORD = 'fin' LOGIN_NAME = 'fin_user'  
DEFAULT_ROLE='finance_user' DEFAULT_WAREHOUSE = 'finance_wh'  
MUST_CHANGE_PASSWORD = FALSE;
```

---CREATING ETL USER

```
CREATE USER etl_user1 PASSWORD = 'etl' LOGIN_NAME = 'etl_user'
DEFAULT_ROLE='etl_developer' DEFAULT_WAREHOUSE = 'elt_wh'
MUST_CHANGE_PASSWORD = FALSE;
```

---ASSIGNING ROLES TO THE USER

```
GRANT ROLE marketing_analyst TO USER mkt_user1;
```

```
GRANT ROLE finance_user TO USER fin_user1;
```

```
GRANT ROLE etl_developer TO USER etl_user1;
```

NAME ↑	DISPLA...	STATUS	LAST L...	TYPE	MFA	OWNER
ETL_USER1	ETL_USE...	• Ena...	—	Person	No	ACCOUNTADMIN
FIN_USER1	FIN_USER1	• Ena...	—	Person	No	ACCOUNTADMIN
KEERTHIKONDURU	KEERTHI...	• Ena...	1 minute ...	—	No	ACCOUNTADMIN
MKT_USER1	MKT_US...	• Ena...	—	Person	No	ACCOUNTADMIN

4.CREATING TABLE AND DATABASE

---DATABASE CREATION

```
create or replace database retail_marketing_db;
```

```
use retail_marketing_db;
```

---SCHEMA CREATION

```
create or replace schema public;
```

```
use schema public;
```

---CREATING TABLE CAMPAIGN SALES

```
create or replace table campaign_sales as
```

```
select c.C_CUSTOMER_SK AS ID,
```

```
c.C_CUSTOMER_ID AS CUSTOMER_ID,
```

```
c.C_FIRST_NAME AS FIRST_NAME,
```

```

c.C_LAST_NAME AS LAST_NAME,

s.ss_item_sk AS item_id,

s.ss_sold_date_sk AS SOLD_sk,

s.ss_sold_time_sk AS SOLD_time_sk,

s.SS_QUANTITY AS QUANTITY,

s.ss_sales_price AS PRICE,

s.ss_ticket_number AS TICKET_NO,

d.d_date as date,

i.i_product_name as product_name,

i.i_category as category,

i.i_class as product_class

FROM SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.STORE_SALES s

JOIN SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.CUSTOMER c

    ON s.ss_customer_sk = c.c_customer_sk

JOIN SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.DATE_DIM d

    ON s.ss_sold_date_sk = d.d_date_sk

JOIN SNOWFLAKE_SAMPLE_DATA.TPCDS_SF10TCL.ITEM i

    ON s.ss_item_sk = i.i_item_sk limit 1000000000;

```

112
113  select * from campaign_sales limit 100;
114

Results (just now)

Table Chart Pivot 🔍 100 rows 28ms

	# ID	# CUSTOMER_ID	# FIRST_NAME	# LAST_NAME	# ITEM_ID	# SOLD_SK	# SOLD_TIME_SK	# QUANTITY	# PRICE	# TICKET_NO
										
	404... 497...	AAAAAABCPK... 15.0% AAAAAACGAD... 12.0% +10 more	Kari 15.0% Timothy 12.0% +10 more	Marquez 15.0% Mitchell 12.0% +10 more	103... 396...	2451...2451...	36713 68290	1 99	0.15 13...	5376... 2377...
2	7168941	AAAAAANKDNGAA	Troy	Wallis	199973	2451906	50524	27	89.53	2099065816
3	7168941	AAAAAANKDNGAA	Troy	Wallis	318155	2451906	50524	34	16.91	2099065816
4	22679045	AAAAAAFAOAKFBA	Mark	Bearden	286922	2451906	40349	46	104.29	1207560153
5	7168941	AAAAAANKDNGAA	Troy	Wallis	201635	2451906	50524	39	6.32	2099065816
6	7168941	AAAAAANKDNGAA	Troy	Wallis	394220	2451906	50524	91	16.92	2099065816
7	7168941	AAAAAANKDNGAA	Troy	Wallis	252937	2451906	50524	57	44.64	2099065816
8	22679045	AAAAAAFAOAKFBA	Mark	Bearden	15889	2451906	40349	71	43.09	1207560153
9	22679045	AAAAAAFAOAKFBA	Mark	Bearden	269645	2451906	40349	32	5.46	1207560153
10	22679045	AAAAAAFAOAKFBA	Mark	Bearden	129811	2451906	40349	39	60.48	1207560153
11	22679045	AAAAAAFAOAKFBA	Mark	Bearden	298310	2451906	40349	36	40.89	1207560153

5.GRANT PERMISSIONS

-----PERMISSION TO DATABASE

GRANT USAGE ON DATABASE retail_marketing_db TO ROLE marketing_analyst;

GRANT USAGE ON DATABASE retail_marketing_db TO ROLE finance_user;

GRANT USAGE ON DATABASE retail_marketing_db TO ROLE etl_developer;

----PERMISSION TO SCHEMA

GRANT USAGE ON SCHEMA public TO ROLE marketing_analyst;

GRANT USAGE ON SCHEMA public TO ROLE finance_user;

GRANT USAGE ON SCHEMA public TO ROLE etl_developer;

-----READ ONLY TO SPECIFIC ROLE

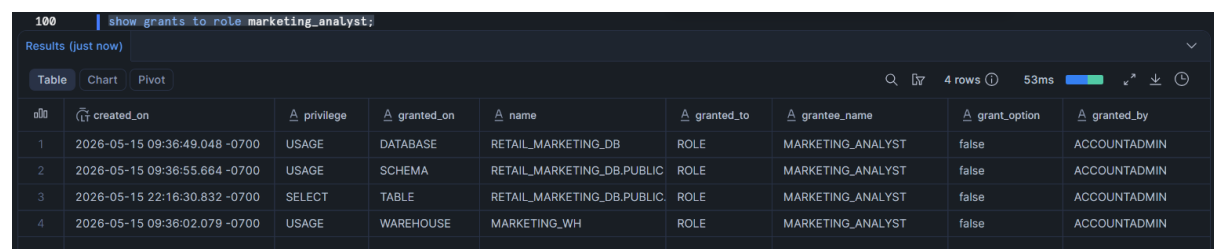
GRANT SELECT ON TABLE campaign_sales TO ROLE marketing_analyst;

GRANT SELECT ON TABLE campaign_sales TO ROLE finance_user;

-----FULL ACCESS TO ETL DEVELOPER

GRANT INSERT,UPDATE,SELECT,DELETE ON TABLE campaign_sales TO ROLE etl_developer;

GRANTS ON ROLE MARKETING_ANALYST



100 | show grants to role marketing_analyst;

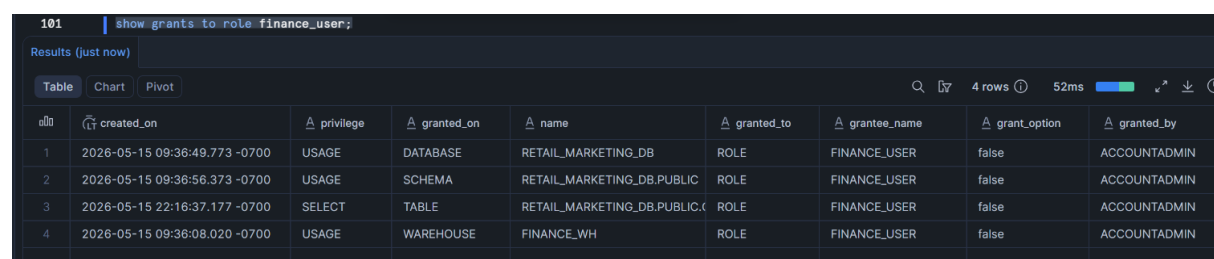
Results (just now)

Table | Chart | Pivot

4 rows | 53ms

#	created_on	privilege	granted_on	name	granted_to	grantee_name	grant_option	granted_by
1	2026-05-15 09:36:49.048 -0700	USAGE	DATABASE	RETAIL_MARKETING_DB	ROLE	MARKETING_ANALYST	false	ACCOUNTADMIN
2	2026-05-15 09:36:55.864 -0700	USAGE	SCHEMA	RETAIL_MARKETING_DB.PUBLIC	ROLE	MARKETING_ANALYST	false	ACCOUNTADMIN
3	2026-05-15 22:16:30.832 -0700	SELECT	TABLE	RETAIL_MARKETING_DB.PUBLIC.C	ROLE	MARKETING_ANALYST	false	ACCOUNTADMIN
4	2026-05-15 09:36:02.079 -0700	USAGE	WAREHOUSE	MARKETING_WH	ROLE	MARKETING_ANALYST	false	ACCOUNTADMIN

GRANTS ON ROLE FINANCE_USER



101 | show grants to role finance_user;

Results (just now)

Table | Chart | Pivot

4 rows | 52ms

#	created_on	privilege	granted_on	name	granted_to	grantee_name	grant_option	granted_by
1	2026-05-15 09:36:49.773 -0700	USAGE	DATABASE	RETAIL_MARKETING_DB	ROLE	FINANCE_USER	false	ACCOUNTADMIN
2	2026-05-15 09:36:56.373 -0700	USAGE	SCHEMA	RETAIL_MARKETING_DB.PUBLIC	ROLE	FINANCE_USER	false	ACCOUNTADMIN
3	2026-05-15 22:16:37.177 -0700	SELECT	TABLE	RETAIL_MARKETING_DB.PUBLIC.C	ROLE	FINANCE_USER	false	ACCOUNTADMIN
4	2026-05-15 09:36:08.020 -0700	USAGE	WAREHOUSE	FINANCE_WH	ROLE	FINANCE_USER	false	ACCOUNTADMIN

GRANTS ON ROLE ETL_DEVELOPER

102 | show grants to role etl_developer;

Results (just now)

	created_on	privilege	granted_on	name	granted_to	grantee_name	grant_option	granted_by
1	2026-05-15 09:36:50.513 -0700	USAGE	DATABASE	RETAIL_MARKETING_DB	ROLE	ETL_DEVELOPER	false	ACCOUNTADMIN
2	2026-05-15 09:36:57.098 -0700	USAGE	SCHEMA	RETAIL_MARKETING_DB.PUBLIC	ROLE	ETL_DEVELOPER	false	ACCOUNTADMIN
3	2026-05-15 22:16:44.960 -0700	DELETE	TABLE	RETAIL_MARKETING_DB.PUBLIC.C	ROLE	ETL_DEVELOPER	false	ACCOUNTADMIN
4	2026-05-15 22:16:44.977 -0700	INSERT	TABLE	RETAIL_MARKETING_DB.PUBLIC.C	ROLE	ETL_DEVELOPER	false	ACCOUNTADMIN
5	2026-05-15 22:16:44.968 -0700	SELECT	TABLE	RETAIL_MARKETING_DB.PUBLIC.C	ROLE	ETL_DEVELOPER	false	ACCOUNTADMIN
6	2026-05-15 22:16:44.972 -0700	UPDATE	TABLE	RETAIL_MARKETING_DB.PUBLIC.C	ROLE	ETL_DEVELOPER	false	ACCOUNTADMIN
7	2026-05-15 09:36:12.596 -0700	USAGE	WAREHOUSE	ELT_WH	ROLE	ETL_DEVELOPER	false	ACCOUNTADMIN

6.PERFORMANCE TESTING

Run a sales analysis query using XSMALL warehouse.

use warehouse marketing_wh;

EXAMPLE 1:

select customer_id,first_name,sum(price) from campaign_sales group by customer_id,first_name;

Workspaces Databases +

Database Explorer

Objects Data Products

Search

Filter

FIRST_DATABASE

SNOWFLAKE

SNOWFLAKE_SAMPLE_DA...

INFORMATION_SCHEMA

TPCDS_SF100TCL

TPCDS_SF101TCL

Tables 24

CALL_CENTER

CATALOG_PAGE

CATALOG_RETURNS

CATALOG_SALES

CUSTOMER

CUSTOMER_ADDRESS

CUSTOMER_DEMOGR...

DATE_DIM

HOUSEHOLD_DEMOG...

My Workspace > day3 > ASSIGNMENT.sql

ACCOUNTADMIN • MARKETING_WH RETAIL_MARKETING_DB PUBLIC Share

```

100 -----performance testing-----
101 use warehouse marketing_wh;
102
103 select * from campaign_sales limit 2;
104
105 select count(*) as products_count,CATEGORY from campaign_sales group by CATEGORY
106 order by products_count;
107
108 select customer_id,first_name,sum(price) from campaign_sales group by customer_id,first_name;

```

Results (just now)

	CUSTOMER_ID	FIRST_NAME	SUM(PRICE)
1	AAAAAAAAEFOPNABA	Kathleen	1194.4
2	AAAAAAAAFPENMABA	Margaret	1788.83
3	AAAAAAAAFNPGBA	Tasha	811.74
4	AAAAAAAAEODNMBCA	Lenora	836.70

EXAMPLE 2:

select * from campaign_sales;

```

114
115 ✦ alter warehouse marketing_wh set warehouse_size = 'xsmall';|
116
117 | select * from campaign_sales;
118
119 -----catche testing-----
120
121 select customer_id,first_name,last_name,item_id,sold_sk,price,date,product_name from campaign_sales order by
    date;

```

Results (6 minutes ago)

Table Chart Pivot 1,000,000,000 rows 5m 50s

#	ID	CUSTOMER_ID	FIRST_NAME	LAST_NAME	ITEM_ID	SOLD_SK	SOLD_TIME_SK
1	42925269	AAAAAAAFNMPOICA	Roger	Williams	189724	2451496	34005
2	35928701	AAAAAAANHKDECCA	Linda	Lacy	350053	2451496	4312
3	4793692	AAAAAAAMFFCJEA	Wade	Thomas	177058	2451496	63437
4	35928701	AAAAAAANHKDECCA	Linda	Lacy	51748	2451496	43124

AFTER CONVERTING THE WAREHOUSE TO SMALL

alter warehouse marketing_wh set warehouse_size = 'small';

EXAMPLE 1:

select * from campaign_sales ;

```

109
110 alter warehouse marketing_wh set warehouse_size = 'small';
111 ✦ | Ctrl+I to generate
112 | select * from campaign_sales;
113
114 alter warehouse marketing_wh set warehouse_size = 'small';
115
116

```

Results (2 minutes ago)

Table Chart Pivot 1,000,000,000 rows 2m 48s

#	ID	CUSTOMER_ID	FIRST_NAME	LAST_NAME	ITEM_ID	SOLD_SK	SOLD_TIME_SK
1	8231740	AAAAAAAMD LJNHAA	Duncan	Cochrane	162405	2452205	38748
2	8231740	AAAAAAAMD LJNHAA	Duncan	Cochrane	25471	2452205	38748
3	8231740	AAAAAAAMD LJNHAA	Duncan	Cochrane	321829	2452205	38748

EXAMPLE 2:

select customer_id,first_name,sum(price) from campaign_sales group by
customer_id,first_name;

```

103 -- select * from campaign_sales limit 2;
104
105 -- select count(*) as products_count,CATEGORY from campaign_sales group by CATEGORY
106 -- order by products_count;
107
108 | select customer_id,first_name,sum(price) from campaign_sales group by customer_id,first_name;
109
110 alter warehouse marketing_wh set warehouse_size = 'small';
111 ✨ | Ctrl+I to generate
112

```

results (just now)

Table Chart Pivot

47,990,407 rows 30ms

	CUSTOMER_ID	FIRST_NAME	SUM(PRICE)
1	AAAAAAAEFOPNABA	Kathleen	1194.4
2	AAAAAAAFPENMABA	Margaret	1788.83
3	AAAAAAAFNPGDBA	Tasha	811.74
4	AAAAAAAEODNMBCA	Lenora	836.70

Feedback

alter warehouse marketing_wh set warehouse_size = 'xsmall';

NOTE:

WITH SMALL SIZE THE WAREHOUSE TOOK ALMOST TOOK '5M50S' BUT WITH WAREHOUSE HAVING SIZE SMALL IT TOOK '2M15S'

7.CATCHE TESTING

BEFORE CATCHEING

select customer_id,first_name,last_name,item_id,sold_sk,price,date,product_name from campaign_sales order by date;

```

117 -----cathe testing-----
118
119 ✨ | select customer_id,first_name,last_name,item_id,sold_sk,price,date,product_name from campaign_sales order by
    | date;|

```

results (5 minutes ago)

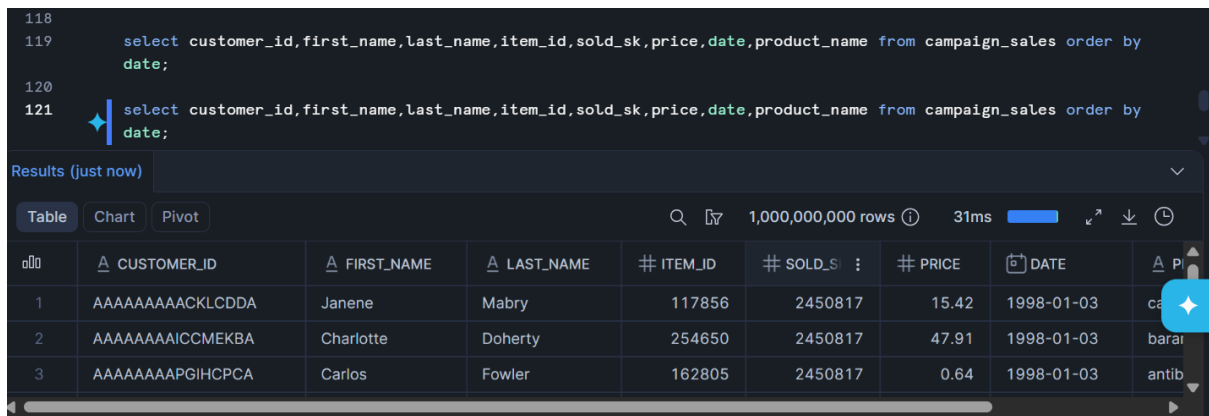
Table Chart Pivot

1,000,000,000 rows 5m 38s

	CUSTOMER_ID	FIRST_NAME	LAST_NAME	ITEM_ID	SOLD_SK	PRICE	DATE	PRODUCT_NAME
1	AAAAAAAACKLCDDA	Janene	Mabry	117856	2450817	15.42	1998-01-03	ca
2	AAAAAAAICCMEKBA	Charlotte	Doherty	254650	2450817	47.91	1998-01-03	barar
3	AAAAAAAPGIHCPCA	Carlos	Fowler	162805	2450817	0.64	1998-01-03	antib

AFTER CATCHEING

select customer_id,first_name,last_name,item_id,sold_sk,price,date,product_name from campaign_sales order by date;



The screenshot shows a SQL query execution interface. The query is: `select customer_id,first_name,last_name,item_id,sold_sk,price,date,product_name from campaign_sales order by date;`. The results are displayed in a table with 9 columns: CUSTOMER_ID, FIRST_NAME, LAST_NAME, ITEM_ID, SOLD_SK, PRICE, DATE, and PRODUCT_NAME. The first three rows are visible, showing data for Janene Mabry, Charlotte Doherty, and Carlos Fowler, all dated 1998-01-03.

	CUSTOMER_ID	FIRST_NAME	LAST_NAME	ITEM_ID	SOLD_SK	PRICE	DATE	PRODUCT_NAME
1	AAAAAAAACKLCDDA	Janene	Mabry	117856	2450817	15.42	1998-01-03	ca
2	AAAAAAAICMEKBA	Charlotte	Doherty	254650	2450817	47.91	1998-01-03	barai
3	AAAAAAAAPGIHCPA	Carlos	Fowler	162805	2450817	0.64	1998-01-03	antib

NOTE:

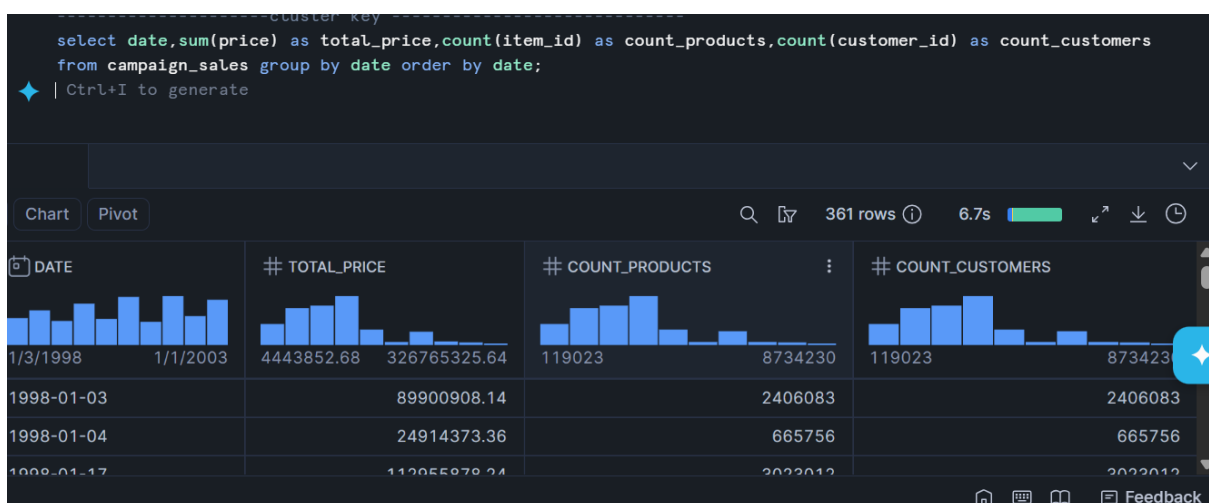
BEFORE CATCHING IT LAKES ALMOST 5MINUTE TO EXUTE THE QUERY BUT SECOND TIME WHEN I EXECUTE IT I JUST TAKE 31MS

8.CLUSTER KEY

first code:

BEFORE CLUSTER KEY

select date,sum(price) as total_price,count(item_id) as count_products,count(customer_id) as count_customers from campaign_sales group by date order by date;



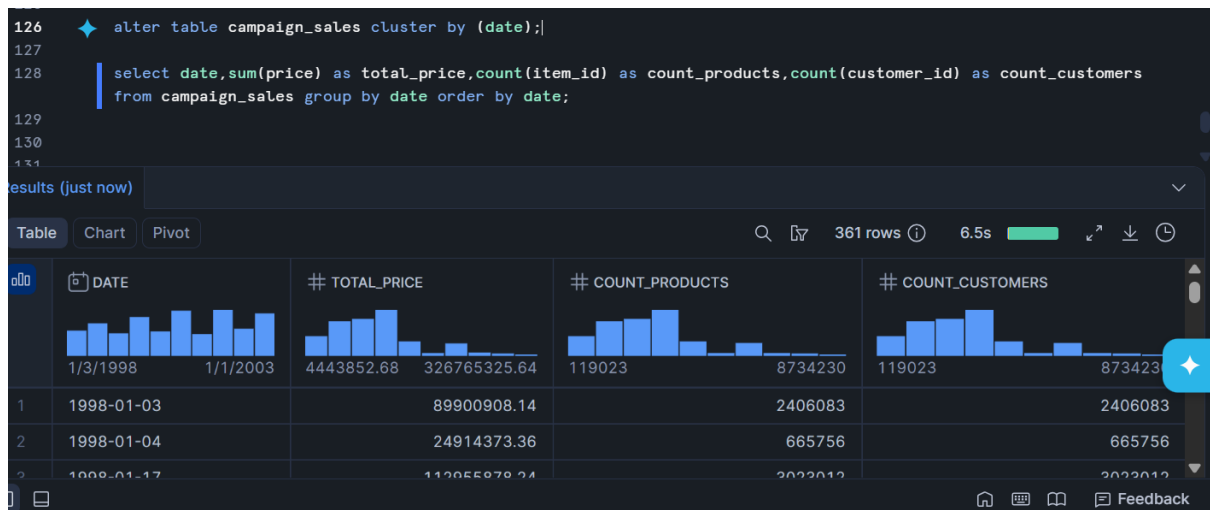
The screenshot shows a SQL query execution interface. The query is: `select date,sum(price) as total_price,count(item_id) as count_products,count(customer_id) as count_customers from campaign_sales group by date order by date;`. The results are displayed in a table with 4 columns: DATE, TOTAL_PRICE, COUNT_PRODUCTS, and COUNT_CUSTOMERS. The first three rows are visible, showing data for 1998-01-03, 1998-01-04, and 1998-01-17.

DATE	TOTAL_PRICE	COUNT_PRODUCTS	COUNT_CUSTOMERS
1998-01-03	89900908.14	2406083	2406083
1998-01-04	24914373.36	665756	665756
1998-01-17	112055979.24	2022012	2022012

AFTER CLUSTER KEY

```
alter table campaign_sales cluster by (date);
```

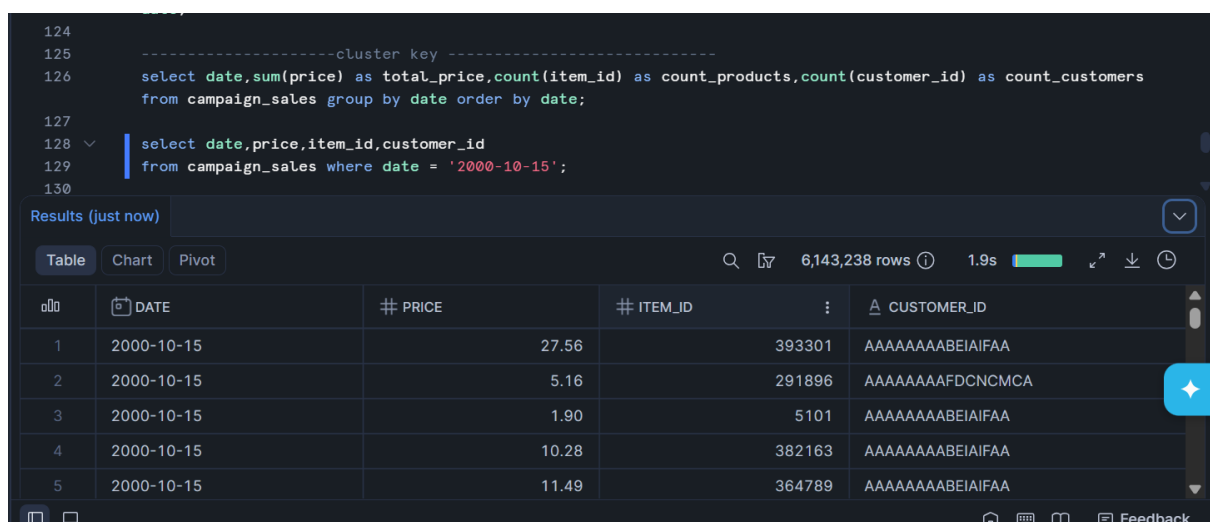
```
select date,sum(price) as total_price,count(item_id) as count_products,count(customer_id) as count_customers from campaign_sales group by date order by date;
```



Second one:

BEFORE ADDING CLUSTER KEY

```
select date,price,item_id,customer_id  
from campaign_sales where date = '2000-10-15';
```

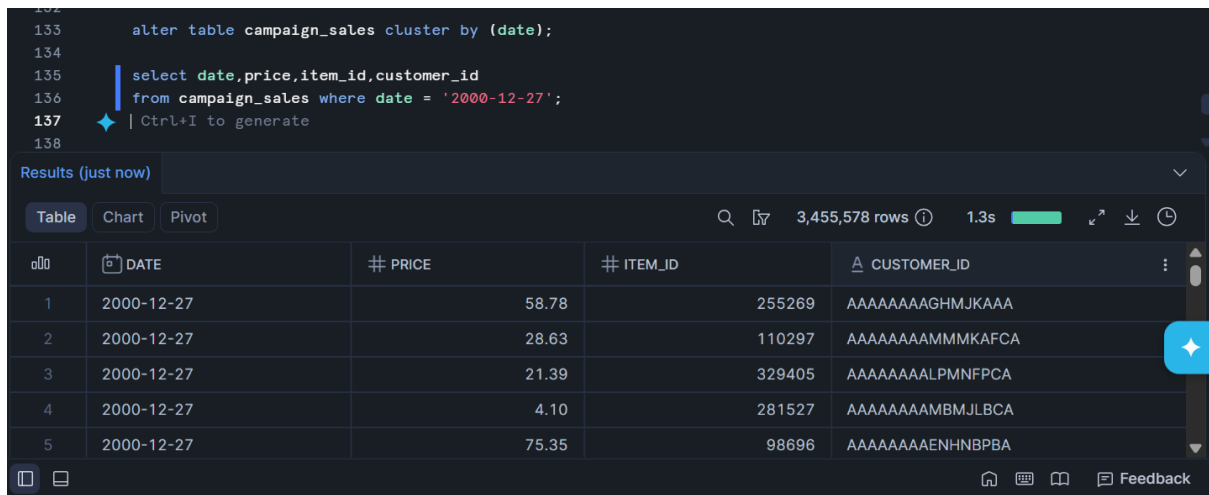


AFTER CREATING CLUSTER KEY

alter table campaign_sales cluster by (date);

select date,price,item_id,customer_id

from campaign_sales where date = '2000-12-27';



Results (just now)

	DATE	PRICE	ITEM_ID	CUSTOMER_ID
1	2000-12-27	58.78	255269	AAAAAAAAAGHMKAAA
2	2000-12-27	28.63	110297	AAAAAAAAAMMMKAFCA
3	2000-12-27	21.39	329405	AAAAAAAAALPMNFPCA
4	2000-12-27	4.10	281527	AAAAAAAAAMBMJLBCA
5	2000-12-27	75.35	98696	AAAAAAAAENHNPBA

NOTE:

USING CLUSTER KEY MAKE QUERY EXECUTION OPTIMIZED AS IT PARTITIONED IN A ORGANIZED MANNER.